



Case Discussions of Stages of gingivitis and periodontal pocket formation

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A large, horizontal, pink brushstroke graphic with a rough, textured edge on the left side and a smoother edge on the right. It serves as a background for the text 'Learning outcomes'.

*Learning
outcomes*

**Describe the features of
different stages of gingivitis**

**Illustrate process of formation
of periodontal pocket**

Healthy Gingiva

- **Clinically healthy gingival tissues appear pink, are not swollen or inflamed, and are firmly attached to the underlying tooth and bone, with minimal/no bleeding on probing**



CLINICAL FEATURES OF HEALTHY GINGIVA

Color: coral pink or salmon pink

Consistency: firm/tight and resilient, well adapted

Contour: scalloped, knife-edge gingival margin, pointed papillae

Surface texture: matte, stippled

Tissue resistance: present to probe penetration

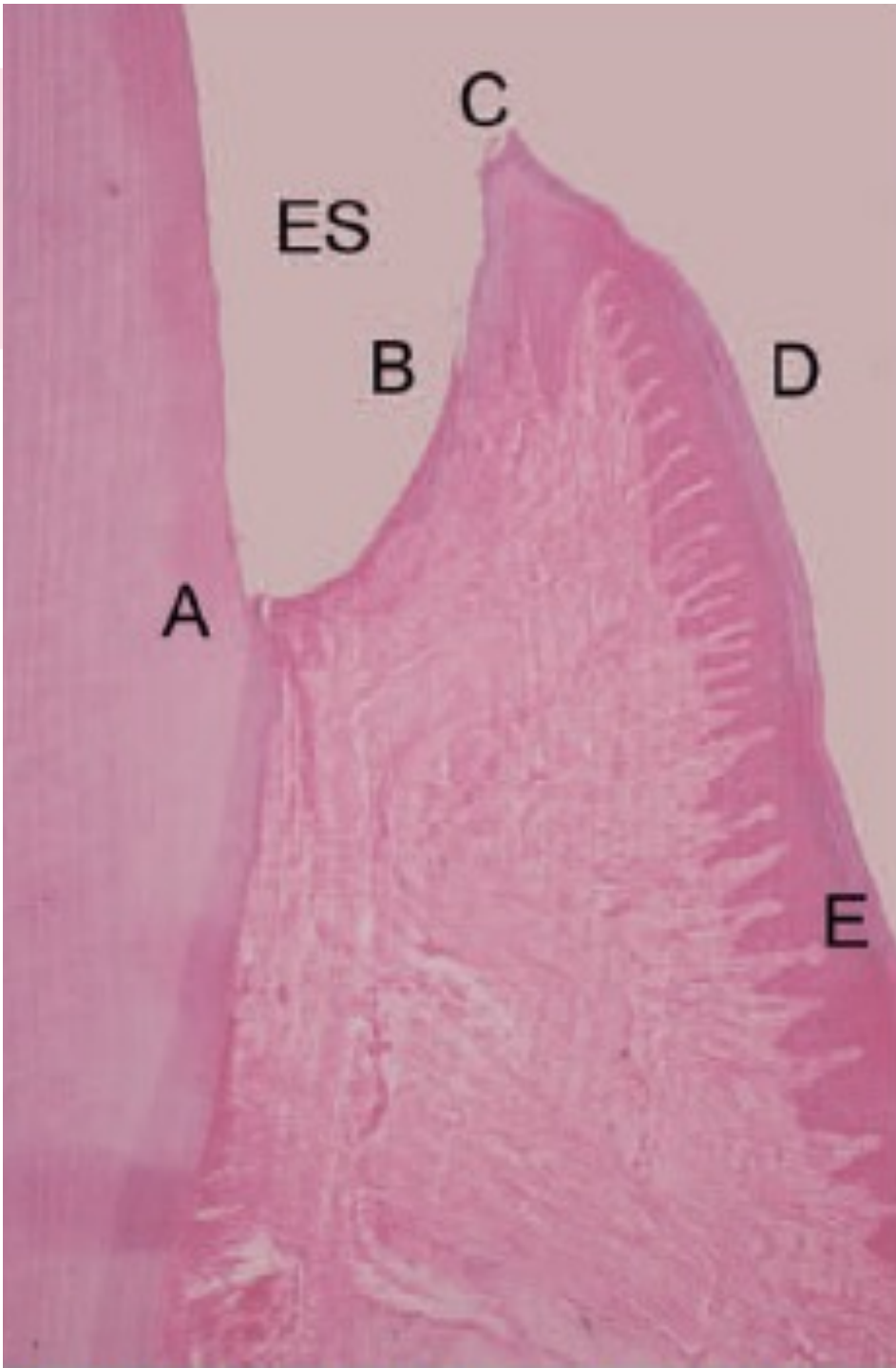
Bleeding on probing: absent or slight

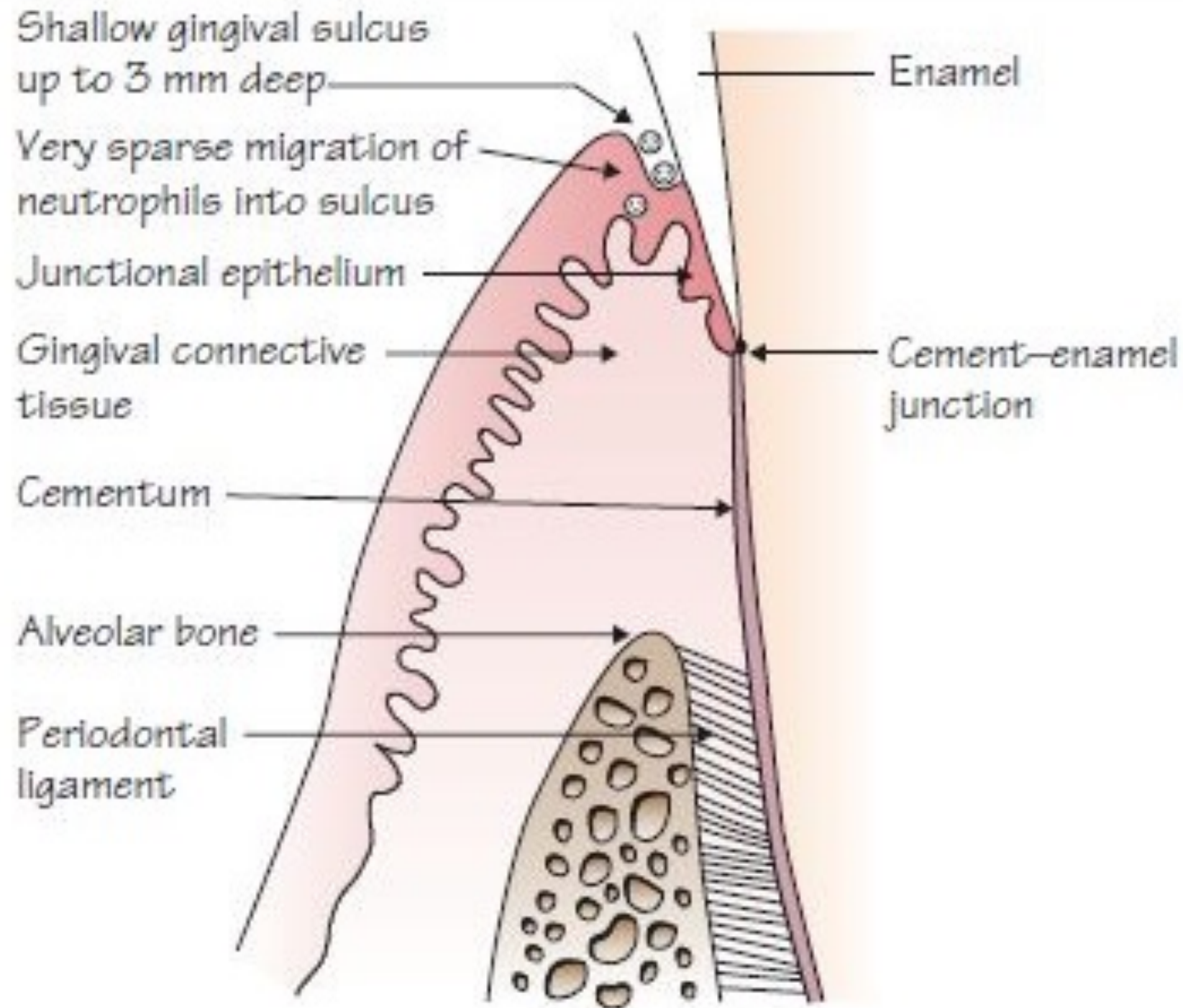
Pain on probing: absent or slight

- In clinically healthy gingiva, connective tissue contains some inflammatory cells, like **neutrophils** in response to the continued presence of bacteria and their products in the gingival sulcus
- In addition, ***lymphocytes and macrophages*** also accumulate
- In clinically healthy tissues, this steady-state equilibrium between ***low-grade inflammation*** in the tissues may persist for many years or lifetime without clinical signs of gingivitis

Histologic appearance of healthy gingiva-- (demineralized tooth with gingival tissues)

- Cementoenamel junction (A)
- Enamel space (ES)
- Gingival health is characterized by the organization of the epithelium into distinct zones; junctional epithelium (A-B), sulcular epithelium (B-C), free gingiva (C-D), and attached gingiva (D-E)
- The gingival connective tissue is composed of densely packed, collagen bundles with few scattered inflammatory cells





**There is very sparse
neutrophil migration
into the sulcus and
no inflammatory
response**

Gingivitis

- **Plaque accumulation → inflammation and clinical signs of gingivitis**
- **Infiltration of the connective tissues by defense cells, neutrophils, macrophages, plasma cells, and lymphocytes**



Extracellular release of their destructive enzymes



Collagen depletion and proliferation of the JE

- ***Vasodilation and increased vascular permeability* -- leakage of fluid and passage of defense cells from the vasculature into the tissues-- enlargement of the tissues, which appear erythematous and edematous**

- Histologic changes that occur in the gingival tissues are described as the *initial, early, established, and advanced gingival lesions*

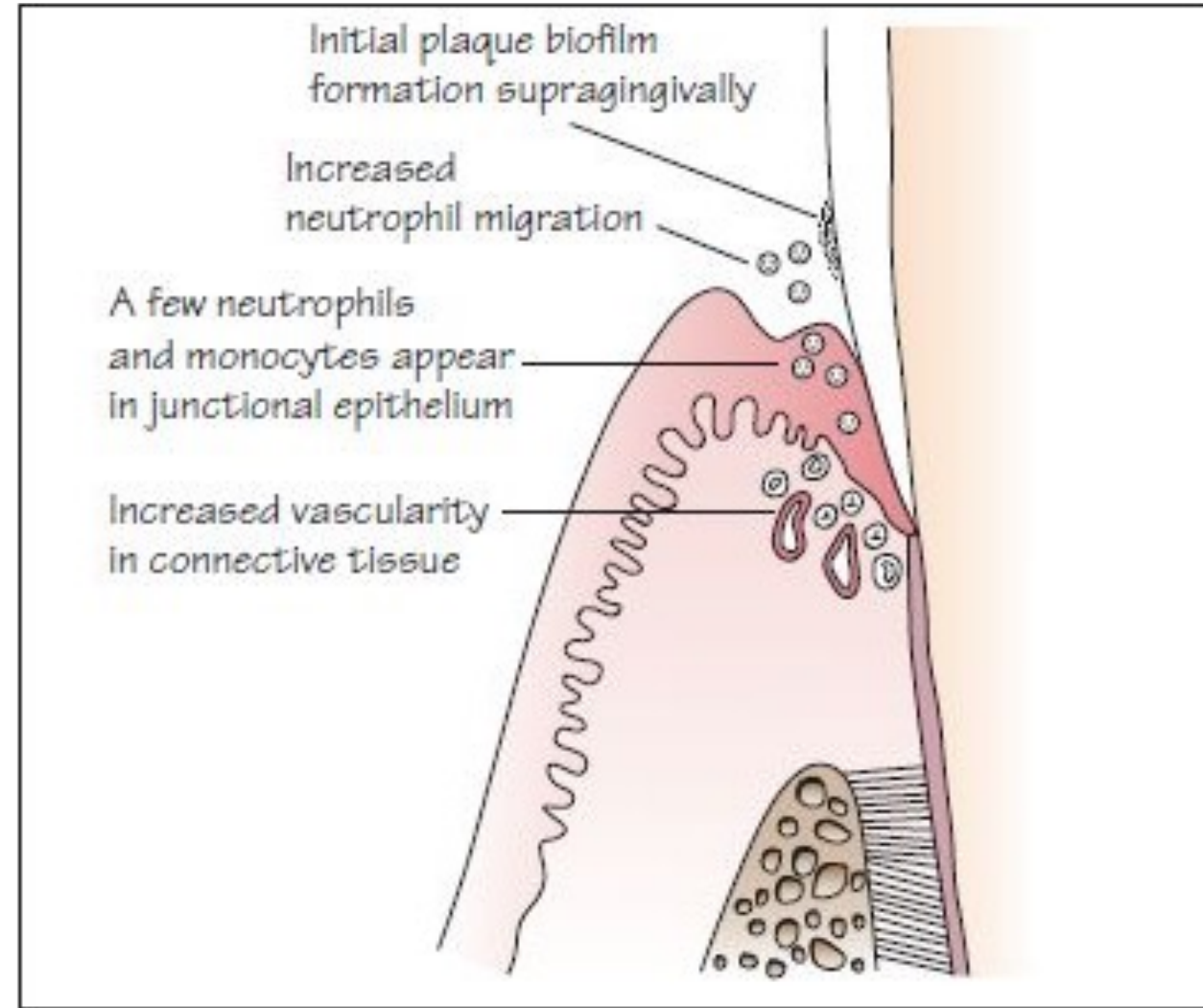
STAGE	TIME (DAYS)	PATHOLOGIC EVENTS	PREDOMINANT IMMUNE CELLS	CLINICAL SIGNS
Initial (Stage I)	2-4	Vascular dilation Vasculitis Infiltration by PMNs into sulcular and junctional epithelium Perivascular loss of collagen	PMNs	Gingival fluid flow
Early (Stage II)	4-7	Vascular proliferation Infiltration by PMNs into sulcular and junctional epithelium T lymphocytes increase Retepeg formation Increased loss of collagen around the infiltrate	Lymphocytes	Erythema Bleeding on probing
Established (Stage III)	14-21	Same as stage II but more advanced Continued loss of collagen	Plasma cells	Erythema Bleeding on probing Changes in size, color, texture etc
Advanced (Stage IV)		Destructive changes extend into bone and other periodontal tissues	Plasma cells	Similar to stage III Attachment loss and bone loss

Stage I gingivitis—Initial lesion

- First manifestations-- *dilated capillaries and increased blood flow*
- Clinically, not apparent
- Stage of *Subclinical Gingivitis*



- **Microscopically, widening of small capillaries or venules in the connective tissue beneath JE and adherence of neutrophils to vessel walls (margination) within 1 week**
- **Leukocytes (PMNs)— leave the capillaries and are seen in the connective tissue, the junctional epithelium, and the gingival sulcus**
- **Exudation of fluid**
- **Perivascular collagen loss**

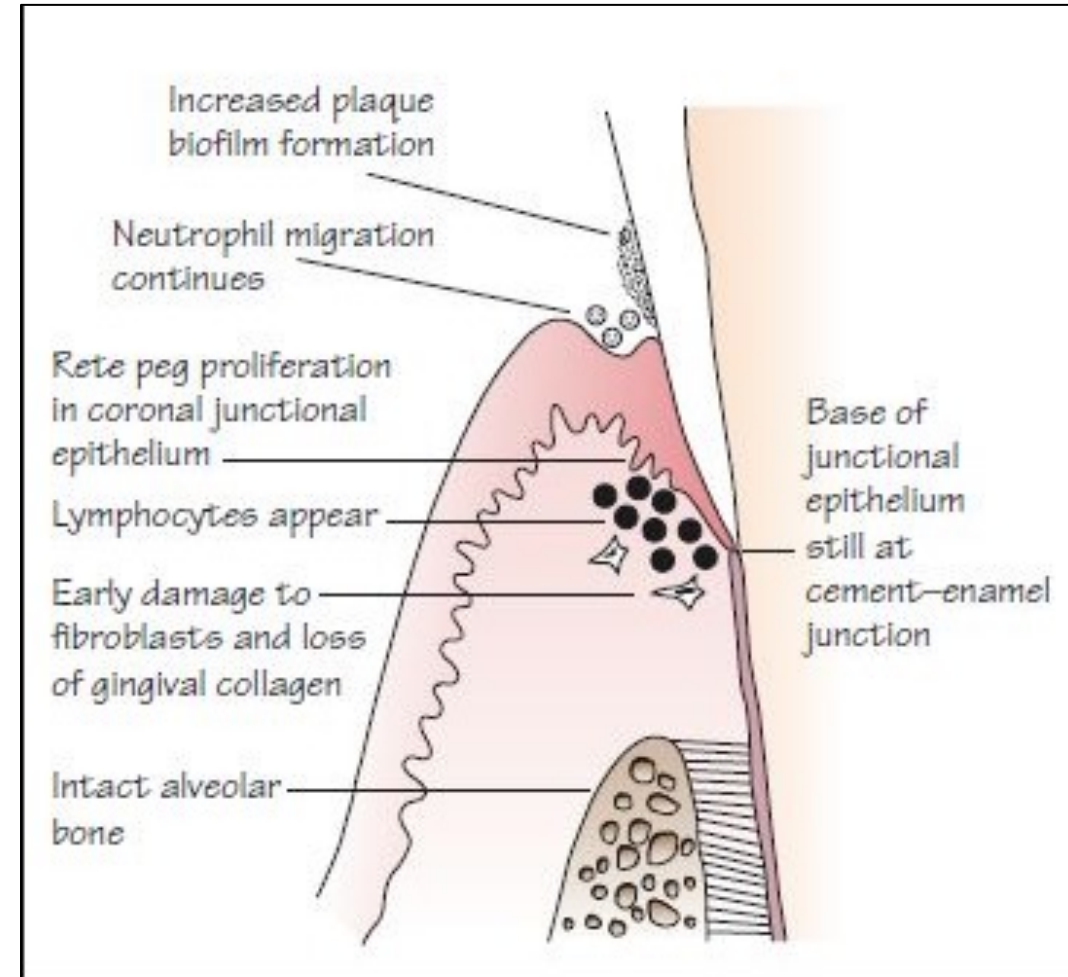


Stage II Gingivitis—Early Lesion

- 4-7 days after plaque accumulation
- Erythema – due to proliferation of capillaries and the increased formation of capillary loops between rete pegs or ridges
- Bleeding on probing
- Gingival fluid flow



- Microscopically, leukocyte infiltration in the connective tissue beneath the JE, mainly lymphocytes
- All changes seen in the initial lesion continue to intensify
- JE becomes densely infiltrated with neutrophils with development of rete pegs
- Increased collagen destruction
- PMNs engulf the bacteria by phagocytosis

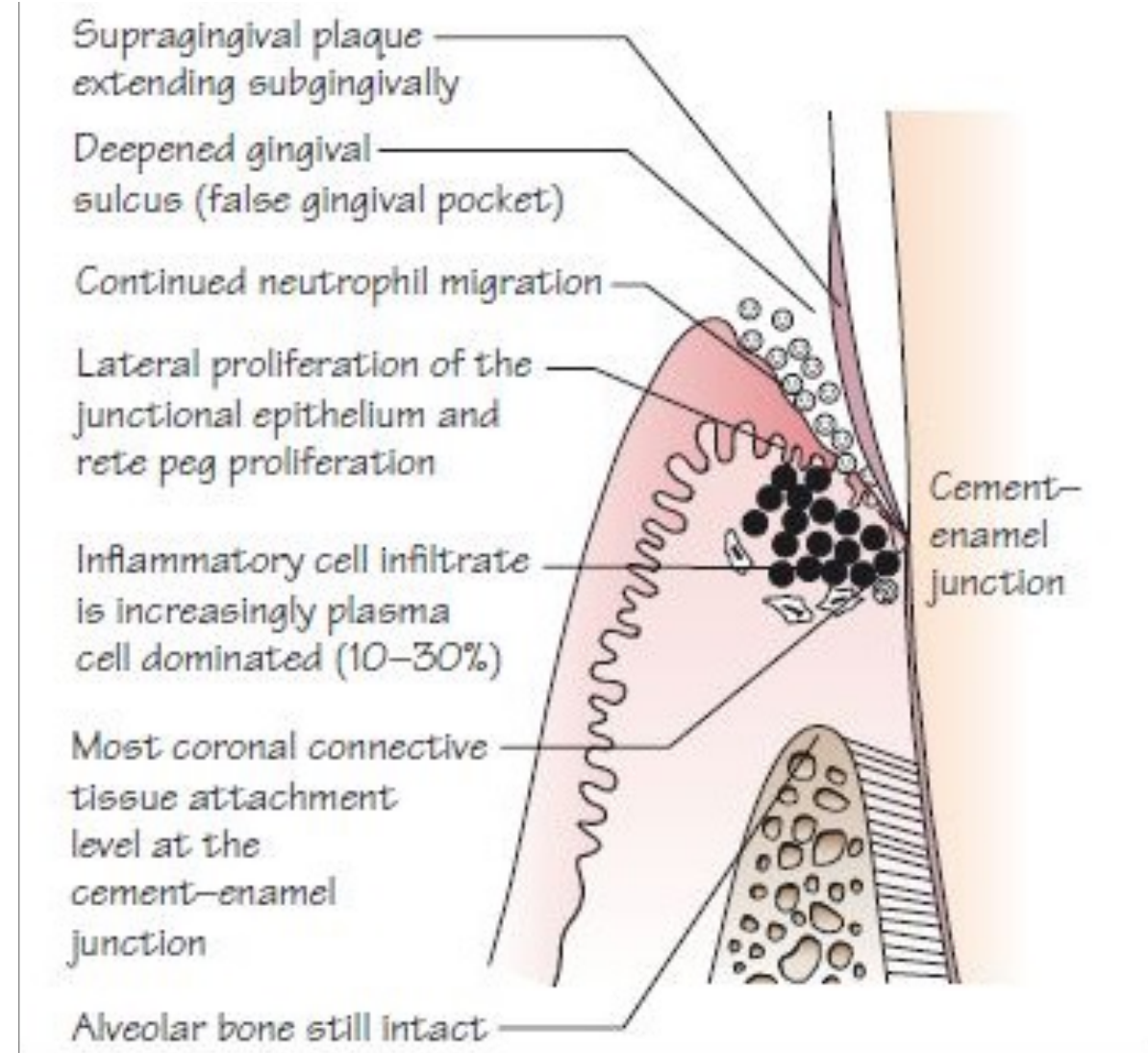


Established lesion

- 2 to 3 weeks after plaque accumulation
- Blood vessels become engorged, congested, venous return is impaired, and the blood flow becomes sluggish --- localized gingival *anoxemia*-- bluish hue on the reddened gingiva
- Extravasation of erythrocytes into the connective tissue and breakdown of hemoglobin into its component pigments can deepen the color of inflamed gingiva



- Increased number of plasma cells--- invade into the connective tissue
- JE epithelium --widened intercellular spaces filled with granular cellular debris, including lysosomes
- JE develops rete pegs that protrude into the connective tissue
- In the connective tissue, collagen fibers are destroyed around the infiltrate of intact and disrupted plasma cells, neutrophils, lymphocytes, monocytes, and mast cells



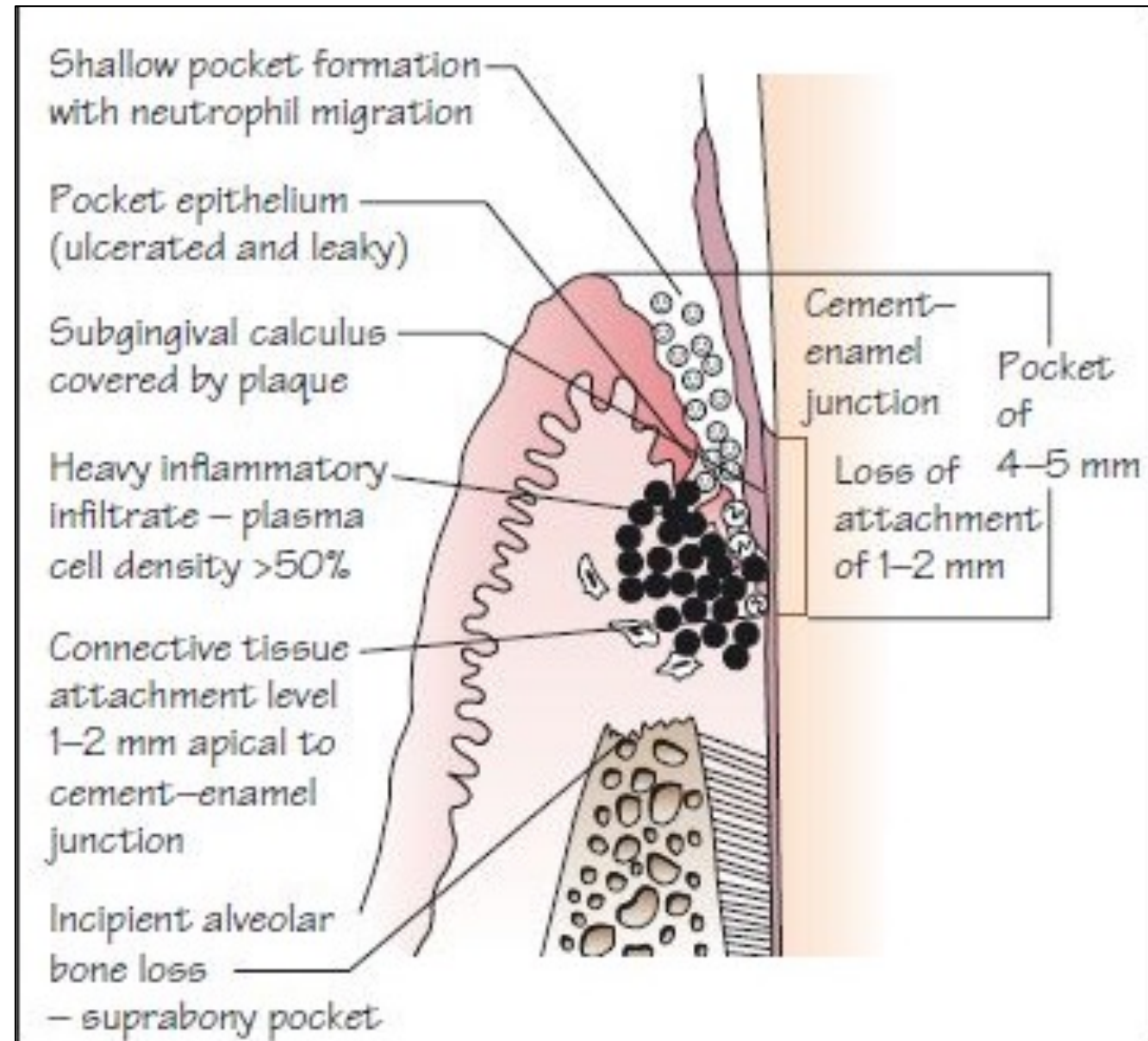
Advanced lesion

- ▶ The extension of the lesion into alveolar bone characterizes the fourth stage, which is known as the ***advanced lesion*** or ***phase of periodontal breakdown***
- ▶ ***Periodontitis—attachment loss and bone loss in addition to all clinical signs of gingivitis***



Patient with periodontitis showing swollen, inflamed gingivae and recession/clinical attachment loss. The gingival margins are rounded and there is loss of contour

- Microscopically, widespread inflammatory and immunopathologic tissue damage
- Plasma cells dominates the connective tissue, and neutrophils continue dominating the junctional epithelium





Healthy Gums



Gingivitis



Periodontitis



Advanced
Periodontitis

CLINICAL FEATURES OF GINGIVITIS

Color: erythematous, cyanotic

Consistency: soft, edematous, spongy, loosely adapted

Contour: bulbous, swollen papillae, rolled/rounded margin

Surface texture: smooth, shiny

Tissue resistance: minimal to probe penetration

Bleeding on probing: moderate to severe

Pain: moderate to severe

COLOR OF GINGIVA



Healthy gingiva



In disease



CONSISTENCY



Firm and resilient



Soft and edematous

CONTOUR



Scalloped, knife-edge gingival margins with pointed interdental papillae



Rounded gingival margins with altered scalloping and bulbous interdental papillae

TEXTURE

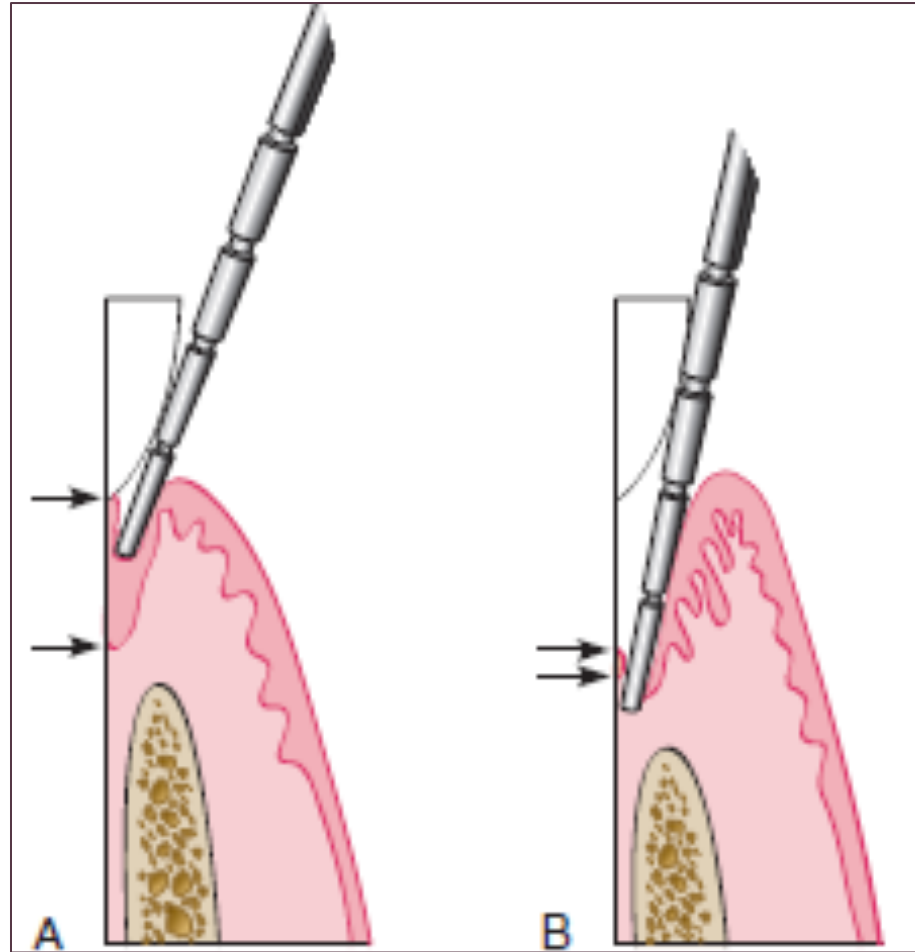


Stippling present



Stippling absent

TISSUE RESISTANCE TO PROBING



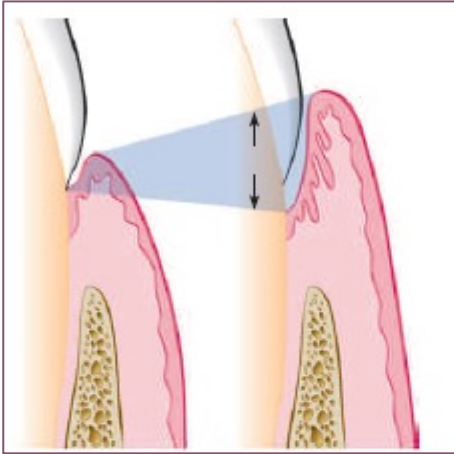
Penetration of probe based on degree of tissue inflammation

BLEEDING ON PROBING

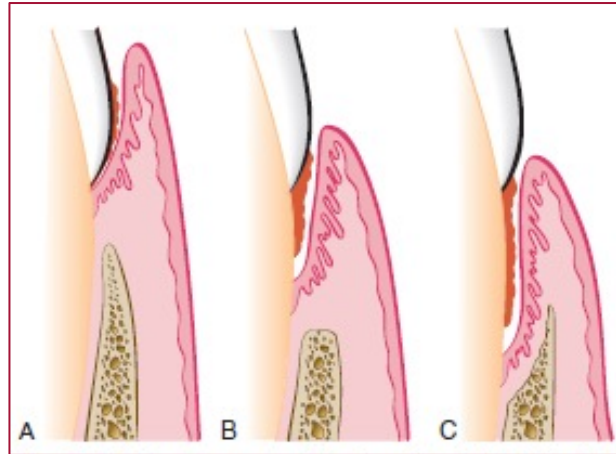


Periodontal Pockets

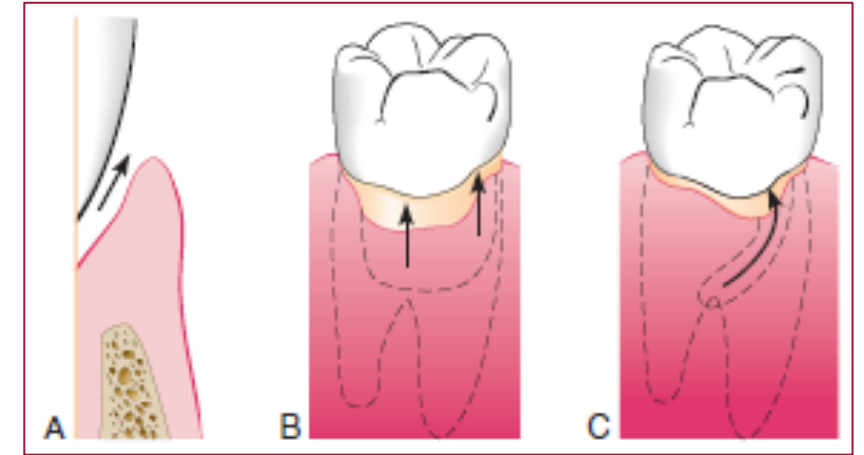
Periodontal pocket--A pathologically deepened gingival sulcus



Pocket formation expands in two directions (arrows)



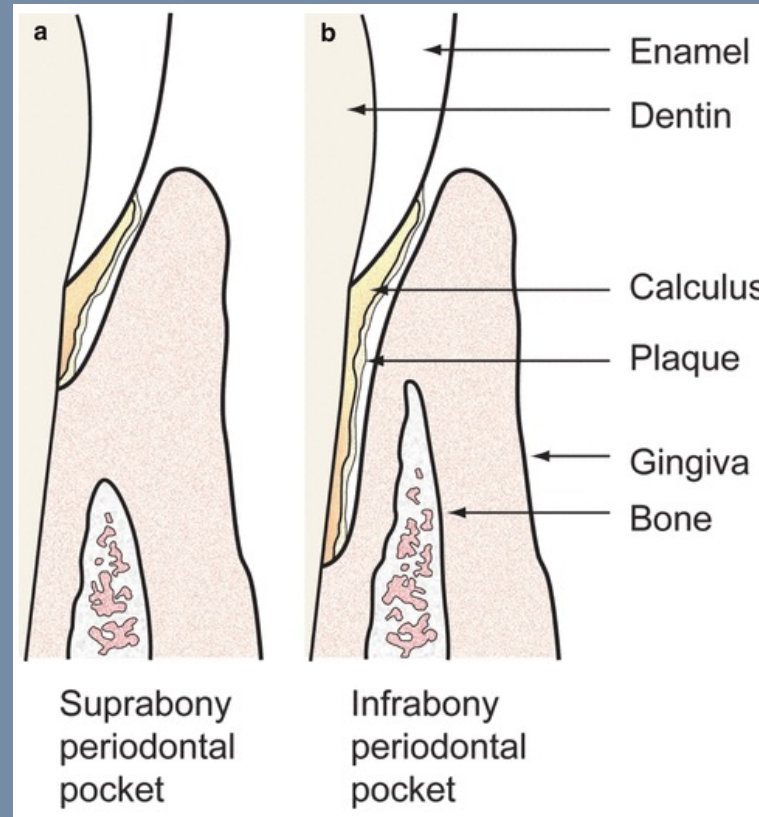
**(A) Gingival pocket.
(B) Suprabony Pocket
(C) Intrabony pocket**



According to involved tooth surfaces

**(A) Simple pocket
(B) Compound pocket
(C) Complex pocket**

Detected by careful exploration with a periodontal probe

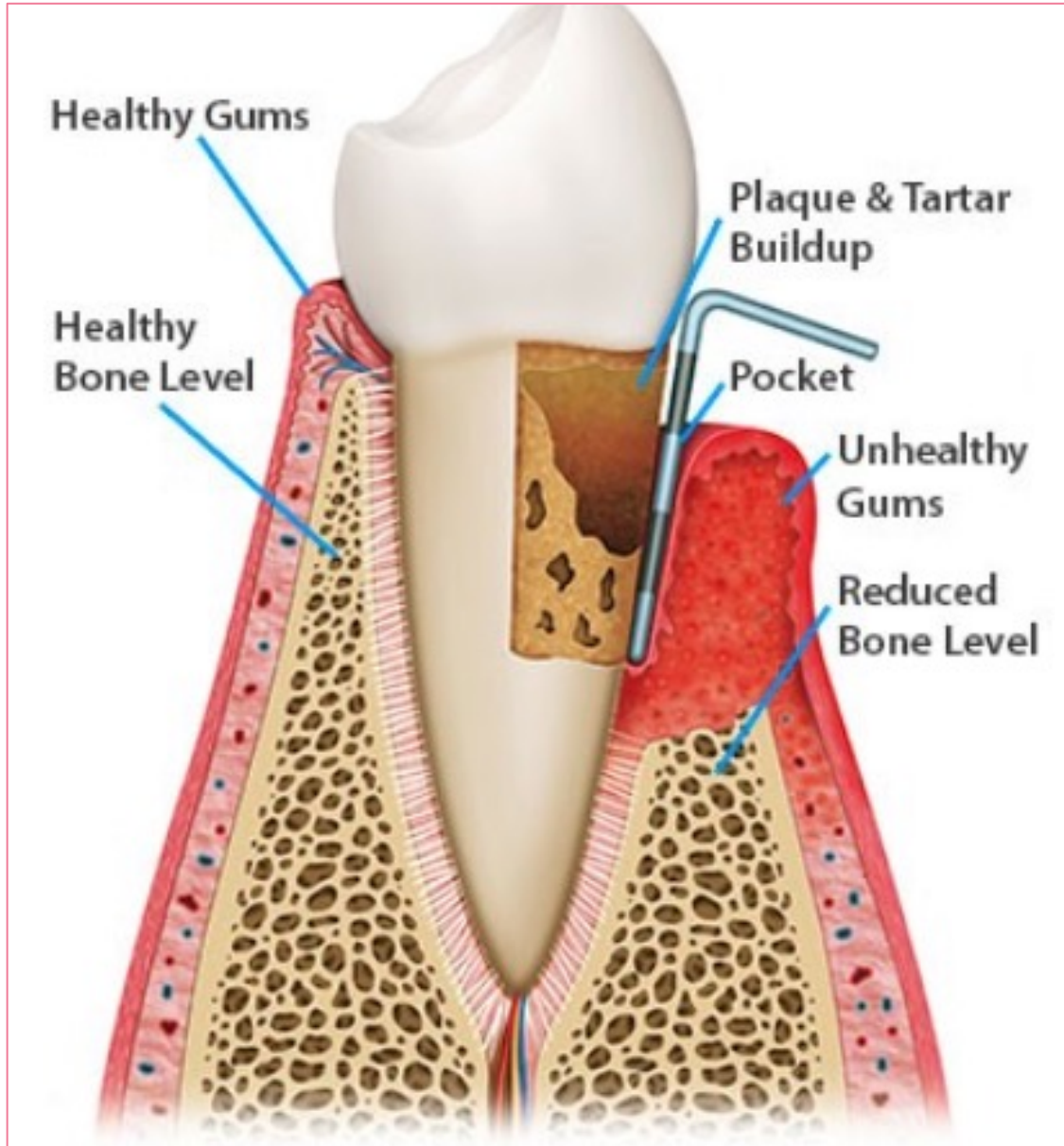


Pattern of destruction of underlying bone

Horizontal—suprabony pocket

Vertical—infrabony pocket

Periodontal pocket



Pathogenesis of pocket formation

Bacterial plaque on tooth surface



Inflammation of marginal gingiva



Deepening of gingival sulcus



Gingival pocket



**Colonization of anaerobic bacteria in Subgingival plaque
(motile rods and spirochetes)**

Mechanisms of collagen loss

PMN

Fibroblast

Collagenase (matrix metalloproteinase)

Phagocytose collagen fibres (by extending cytoplasmic process into PDL-cementum interface)

Destruction of junctional epithelium

Proliferation of epithelial cells along root surface apically

Migration of PMNs to coronal portion of JE (volume of PMNs more than 60%)



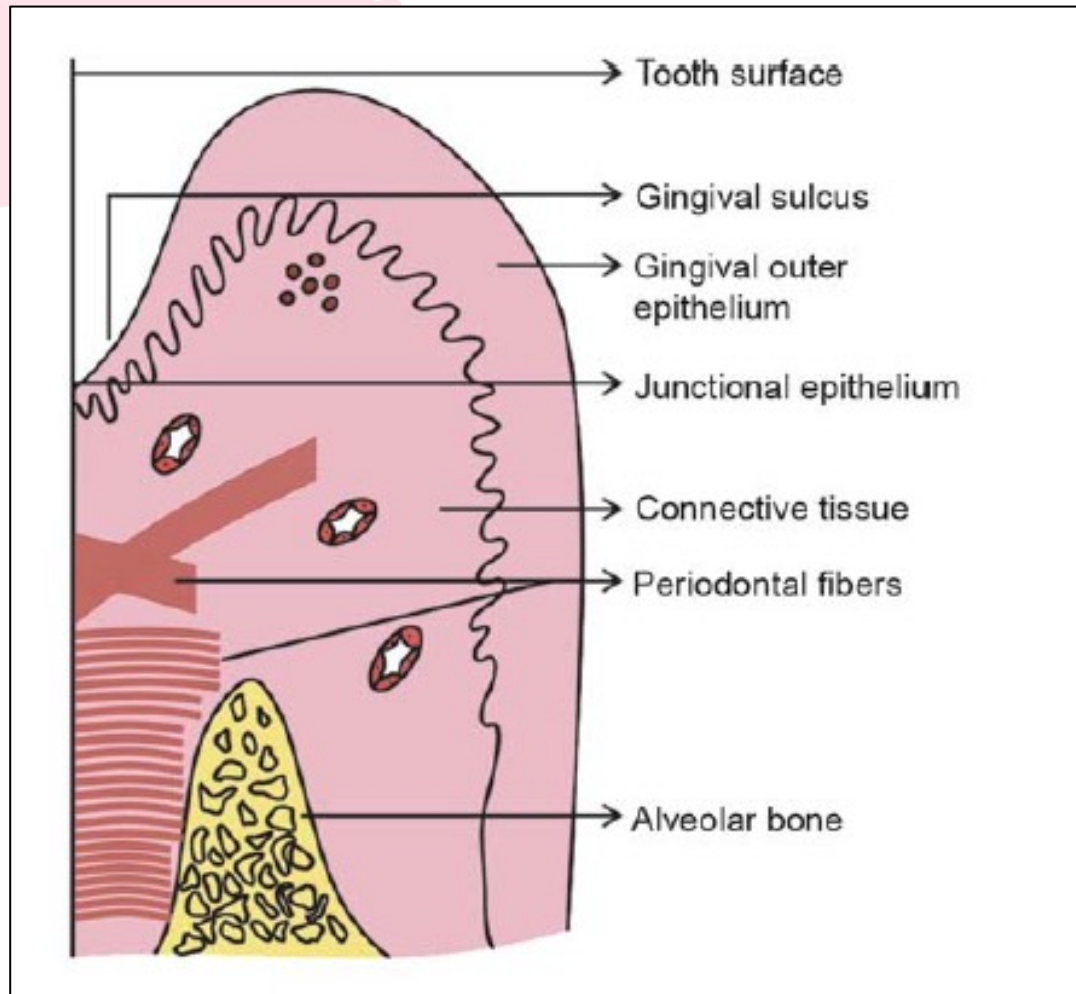
Tissue loses cohesiveness



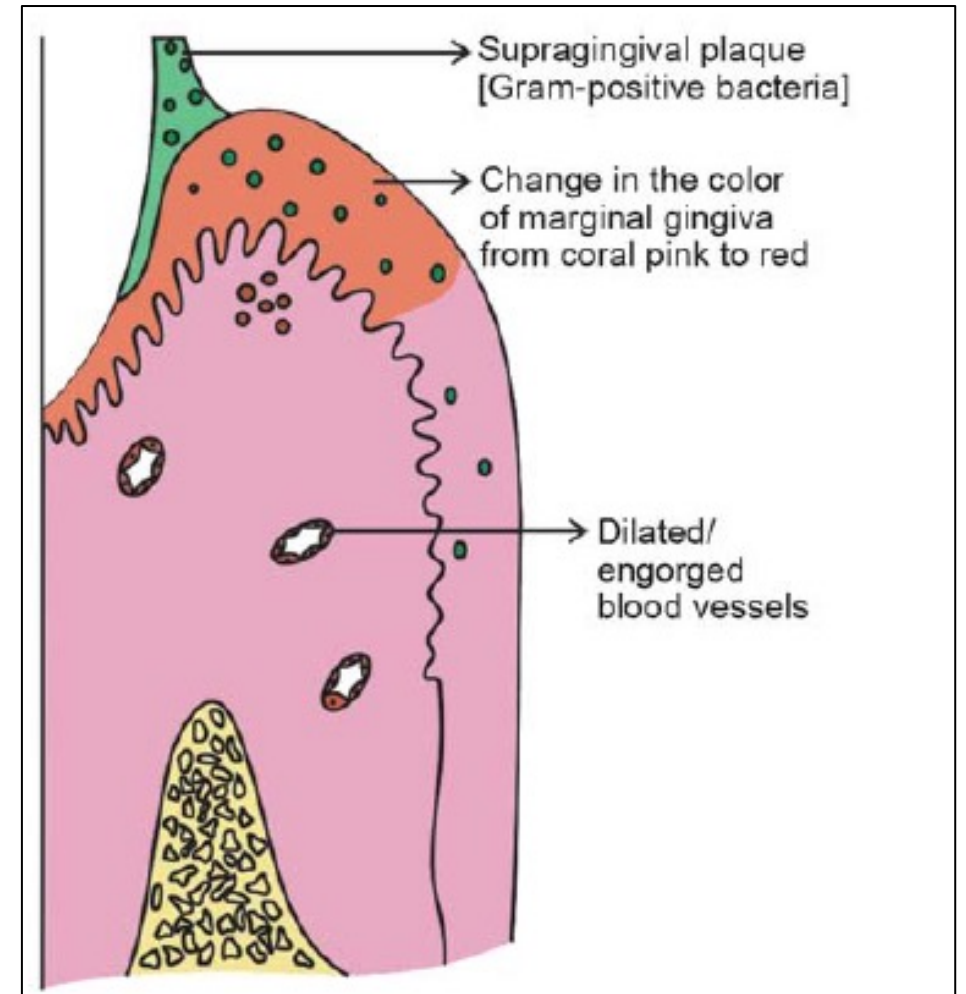
Coronal portion of the junctional epithelium detaches from the root as the apical portion migrates further apically



Periodontal pocket formation

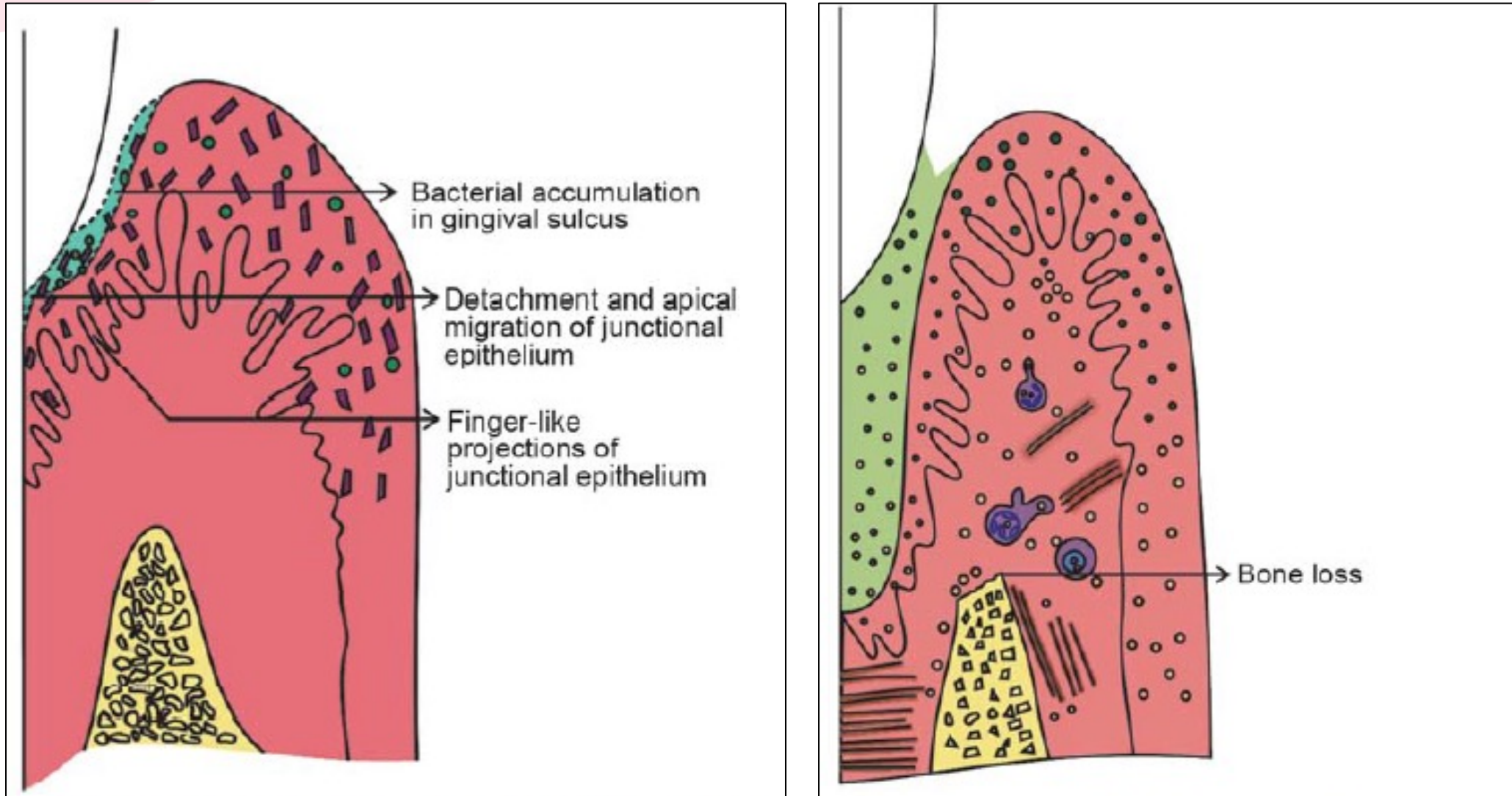


Normal gingiva



Gingivitis

Pocket formation



- **Gingival sulcus**--space between the neck of the tooth and gingival tissue
- **Periodontal pocket**--sulcus, when it deepens due to the apical migration of JE, accompanied by attachment loss
- **Clinical signs of periodontal pockets**—
 - bluish red thickened marginal gingiva,
 - gingival bleeding and suppuration,
 - tooth mobility, diastema formation,
 - localized pain or pain “deep in the bone”
- The only reliable method of locating periodontal pockets and determining their extent is **careful probing of the gingival margin along each tooth surface**

Thank you